Infrastructure planning

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Uncertainty Modelling and Scenario Discovery

Project Sheet 3 (Weeks 5 and 6)

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0 Purpose and scope of Phase 3

During Phase 2, you translated the planning problem into a deterministic system model, defined infrastructure stages, and combined them into deployment plans. Every numerical result was calculated under one fixed set of assumptions. In Phase 3, you replace that single future with a wider set of uncertain scenarios and examine how the resulting spread affects physical transport performance.

The objective is to understand:

- how scenario uncertainties are defined, selected, and represented in the modeling framework;
- how transport-modelling uncertainties differ from post-modelling inputs and from parameters reserved for sensitivity testing;
- why the supplied Gaussian-process surrogate is used, how it is validated, how and when it must be rebuilt or refreshed;
- how Monte Carlo, Latin Hypercube, and Sobol designs sample the uncertainty space;
- how uncertain inputs propagate into selectable Year-1 and Year-40 physical indicators;
- which uncertain drivers matter most for selected outcomes;
- how baseline, static1, and static2 perform under identical scenario draws; and
- under which combinations of uncertain inputs a configurable performance rule is violated.

The accompanying 03_uncertainty_modeling.ipynb notebook guides you through these components. This is the first phase in which the model is evaluated repeatedly across an uncertainty space rather than once at nominal parameter values.

The analysis still uses two fixed snapshots: Year 1 and Year 40. It does not simulate every intermediate year, activate adaptive decision rules, calculate discounted lifecycle costs, or select a preferred plan. Full trajectories and adaptive planning are addressed in Phase 4; monetary appraisal is addressed in Phase 5.

0.1 How to use the Project Sheets

The present exercise sheet provides a short introduction to the course and specific information for the third phase of the semester (Weeks 5-6).

The previous (Weeks 1-4) and subsequent phases (Weeks 7–11) are supported by dedicated project sheets.

The project sheets combine conceptual tasks that help students develop the structure of their final report with practical analyses implemented in the accompanying **Jupyter Notebooks (.ipynb)** provided in the GitHub repository.

Throughout the exercise sheets, tasks are grouped into different categories to clarify what is expected from all groups and to provide opportunities for deeper exploration.

We distinguish between:

Supporting Task (optional) Designed to help you understand the uncertainty framework, become familiar with the Jupyter notebooks, and develop an intuition for the rationale behind each analysis. These tasks do not directly produce outputs for the final report. Especially for students with limited experience in probabilistic modelling or Python, we strongly recommend completing them before interpreting the mandatory results.
Mandatory Task Specifying core activities and deliverables that every group is expected to complete and include in the final report.
Advanced Task (optional) Suggesting extensions for groups interested in exploring additional uncertain parameters, higher-dimensional sampling, or formal global sensitivity analysis.

The exercises generally follow this structure: first defining and visualising the uncertainty assumptions, then sampling and exploring their consequences, and finally identifying sensitivity and vulnerability. Consistency checks should be completed before moving from one part to the next. The exercises will not count as part of the grading. They provide an opportunity to get feedback on your work. You can submit your results (e.g., figures and tables) on Moodle and/or provide access to the lecturers and assistants to your Fork before the next Project sheet is released. You are expected to receive feedback within a week.

1 What uncertain inputs change the transport model?

1.1 Modeling the uncertainties

This section identifies which assumptions may change in the future, and how uncertain those future conditions might be.

The model can vary only assumptions that already have a recognised parameter name. Their default values are stored in code/parameters.py. In the same file, STRUCTURAL_UNCERTAINTIES describes how strongly each selected assumption may vary.

Each uncertain input has two labels that explain how it is used, and their category shows where the input has an effect: “transport_modelling” inputs change demand, mode choice, or traffic assignment, while “post_modelling” inputs are used only in later calculations (such as appraisal).

The “use” label shows why the input is varied: use="scenarios" represents a future condition explored in this notebook, while use="sensitivity" is reserved for a separate sensitivity test in Notebook 05.

Some inputs change over time and others remain constant. A transient input normally starts from today's value and moves gradually towards a sampled value in Year 40. Its Year-1 value may also be sampled when this is explicitly defined. A non-transient input uses the same sampled value throughout the whole 40-year period (or, as in this case, for the two snapshots).

In the supplied example, nine inputs are sampled as scenarios. Four are transport inputs (transport_modelling): PASSENGER_DEMAND_GROWTH, PT_ASC_SHIFT, EBIKE_SHARE and ROAD_FREIGHT_GROWTH. Five are post-modelling inputs: car CO_2 per vehicle-km, the OPEX and CAPEX multipliers, and the accident and local-air-pollution cost multipliers. The Setup cell prints this split (for example, '9 uncertainties (4 transport + 5 post-modelling)').

In the setup, SCENARIO_UNCERTAINTIES chooses which registered scenario inputs are varied. Passenger growth scales the retained OD matrix. PT ASC shift modifies the preference towards PT. Freight growth scales prepared background vehicles on its own. Ebike-share changes cycling travel times. Appraisal inputs, such as cost multipliers and emission factors, are “post_modelling”. Any input that is not selected remains in the model on its default path.

Mandatory Task

Define and justify the uncertainty set used for your case study.

- Check the default values and the registered uncertainty definitions before changing the selection in the setup.
- For every selected input, document what it means, where it acts in the model, whether it changes over time, its plausible range or distribution, and its units.
- Explain why each selected input is an uncertain future condition, rather than an infrastructure decision or an assumption used only for sensitivity testing.
- Confirm that omitted supported inputs follow their nominal paths.Inspect the nominal parameter and uncertainty registries before changing the setup selection.

2 Part 2: Register uncertainty quantiles and map them to trajectories

2.1 Connecting the uncertainties to the EMA Workbench

This section explains how the notebook turns a sampled position on a low-to-high scale into a real model value over time.

The EMA Workbench first chooses a number between 0 and 1: 0 represents the low end of the selected distribution, 0.5 its middle, and 1 its high end. The notebook calls this number u. The helper “get_trajectory” then converts it into the actual values used from Year 1 to Year 40. The middle of a

distribution is not always the same as the default model value, so a middle draw should not automatically be read as the nominal future.

This sampled position is a technical sampling coordinate, not a policy choice or the probability that a future will occur.

Mandatory Task

Connect the selected uncertainties to the EMA Workbench model.

- Register all selected scenario keys as quantile inputs and verify that the wrapper returns the chosen Year-1 and Year-40 outcomes.
- Explain separately what sampling and the transient flag do.
- Check the registry table for fixed versus sampled Year-1 values and for bounded versus unbounded endpoint distributions.
- Do not interpret the sampled quantile as a policy choice or as the probability that a specific future will occur.

2.2 From today's baseline to uncertain Year 40

Here you check whether the selected low, middle, and high Year-40 values form a credible range. Inputs that change over time normally share today's starting value and then move towards different sampled Year-40 values. If an input has fixed minimum and maximum values, the notebook uses them. If it has no hard limits, the notebook uses a wide but practical range and leaves out only extremely unlikely values. The exact statistical settings remain visible in the notebook.

Tutorial/supporting task

Use the registry table to challenge the uncertainty assumptions.

- Compare the default value, the middle sampled value, and the low-to-high Year-40 range for each selected input.
- Identify any range, distribution shape, or Year-1 treatment that is not credible for your project and should be adapted.
- After changing parameter definitions, restart the notebook's running session (the kernel) and run the notebook again so that the new settings are used everywhere.

2.3 Visualising transport uncertainty

The visualisation makes it easier to see which inputs are fixed in Year 1 and how widely they can differ by Year 40. Use the supplied visualisation to compare selected transport inputs in Year 1 and Year 40. Interpret the displayed values in physical units and distinguish fixed starting values from sampled endpoints.

3 Part 3: The fast transport surrogate model

3.1 Why use a surrogate?

A surrogate model is a fast approximation of the full transport model. It is used because running the full model for thousands of possible futures would take too much time.

In the provided example, a Gaussian-process (GP) surrogate learns how congestion delays change for different inputs and for four infrastructure configurations: baseline, Stage 1, Stage 2, and a configuration in which both stages are applied. It predicts delays between origins and destinations by understanding patterns that emerge from the execution of the underlying transport model (see Notebook 02, and the 4-step modeling approach). A predefined response table stores the resulting indicators so they can be evaluated quickly in the later analyses.

The supplied example already includes a trained surrogate and response table. Leave REBUILD_TRANSPORT_SURROGATE and REFRESH_RESPONSE_TABLE set to False if you want to explore the prepared files that are meant to represent the Mehrspur example, or when you want to inspect results without changing the underlying transport model.

For your own case study, rebuild the surrogate based on your network, infrastructure stages, transport inputs, their uncertainty ranges, and assignment settings using REBUILD_TRANSPORT_SURROGATE = True. Rebuilding may take several hours.

Mandatory Task

Understand the structure of the Gaussian Process and rebuild it for your own case study.

- Specify the inputs that are needed to rebuild the surrogate, such as the number of parallel training workers, and the amount of cases that are evaluated through the typical MSA assignment procedure.

3.2 How accurate is the OD-delay approximation?

Before using the surrogate results, check how closely it matches full-model results that were kept separate from training. Read the error table, the observed-versus-predicted plot, the residual plot, and any convergence warnings together. The errors are reported in minutes. Busier origin-destination relations receive more weight because errors there affect more travellers. A residual is the surrogate prediction minus the full-model result. A positive residual means the surrogate overestimates the delay. A small average error can still hide a difficult route pair or a systematic difference between a project and the baseline.

Mandatory Task

- After a rebuild, inspect the saved diagnostics. Check errors separately for each infrastructure configuration and note any warning about accuracy or whether the traffic calculation fully converged.
- Distinguish errors in the surrogate itself from errors introduced when stored results are looked up and from limitations in the original model or data.
- If the notebook prints a validation warning (OD-delay accuracy or assignment convergence), decide if an improvement in the surrogate is needed. For example, increase N_TRAIN, or set ADAPTIVE_FILL_IN = True. Fill-in adds up to MAX_FILL_IN_POINTS extra training runs until the target in FILL_IN_TARGETS is reached (default: demand-weighted RMSE ≤ 0.20 min).

3.3 Inspect an OD relation

You can try to understand which OD-pairs are contributing most to the demand-weighted error calculations.

This might be relevant for your case study area, since a lack of accuracy on specific links and OD-relationships might have a significant impact on your evaluation.

Advanced Task (optional)

This optional check helps you see whether a specific origin-destination pair contributes disproportionately to the surrogate error. Inspect a locally important OD-delay error.

- Use the ranked contribution table to select an OD pair with a material demand-weighted error.

- Explain why local error can matter even when the network-average error is small.

3.4 Keep the evidence and its limits visible

Good surrogate accuracy does not prove that every other part of the transport model is correct. Use the surrogate only within the range of input values on which it was trained; predictions outside that range are less reliable. Also state clearly what the validation does not test: possible error when the response table looks up stored results, incomplete traffic convergence, missing network elements, and incorrect capacity or crowding assumptions.

3.5 From OD delays to reported outcomes

The surrogate predicts transport conditions, while the response table stores the indicators calculated from those conditions. This makes it possible to evaluate many scenarios quickly. If a new indicator can be calculated from information already stored in the table, the GP does not need to be trained again. If the indicator needs a transport quantity that was not stored, the response table must be refreshed. Adding only a new name or label cannot create a result that the model never calculated.

3.6 Minimal worked example: two selected inputs

In order to inspect the data, we can use a simplified example only considers two uncertain inputs. This is used to show clearly how futures are sampled and how they affect the selected outcomes. It is intended as a visual explanation and does not replace the full uncertainty model.

Tutorial/supporting task

The provided example shows sampling and outcome patterns for the Mehrspur case. Use the two-input slice to make sampling coverage and outcome patterns visible. You can adapt the inputs to match your own case study's uncertain parameters.

3.7 Full uncertainty model: all selected scenario inputs

The full model includes all scenario uncertainties selected in the notebook and produces the Year-1 and Year-40 outcomes used in the remaining analyses. As mentioned before, transport inputs affect the surrogate calculations, while post-modelling inputs are applied afterwards. Inputs reserved for sensitivity testing remain at their default values.

Mandatory Task

This is the complete uncertainty model used for the main analyses, with every selected scenario input included. Build and verify the full EMA model used in the remaining analyses.

- Include every selected scenario uncertainty and the chosen physical outcomes for Year 1 and Year 40.
- Check that transport inputs affect the surrogate and response table, while post-modelling inputs are applied only after the transport result has been calculated.
- Keep use="sensitivity" assumptions nominal in this notebook; they are tested in Notebook 05.

4 Part 4: How we generate future scenarios

4.1 Compare sampling designs in a visible 2D example

Sampling methods differ in how they spread model runs across the range of possible futures. Monte Carlo places points randomly and may leave gaps or clusters. Latin Hypercube Sampling (LHS) spreads points across the range of each input, although combinations of inputs may still be uneven. Sobol uses a more structured pattern designed for sensitivity analysis and adds the extra runs needed to study interactions. The notebook calculates the required number of runs automatically. The important point is that more uncertain inputs, especially in a Sobol analysis, require more model evaluations.

Tutorial/supporting task

Compare sampling behaviour in the visible two-input model.

- Plot Monte Carlo, LHS, and Sobol samples on the same axes and compare where they leave gaps, create clusters, or spread points evenly.
- Explain why LHS can spread values evenly along each individual input without covering every combination of two inputs evenly.
- Remember that Sobol includes extra runs for sensitivity calculations, so its methods should not be compared with Monte Carlo or LHS only by counting rows.

4.2 Does the estimated outcome mean stabilize?

The number of sampled futures should be large enough to produce reasonably stable results. The running-average plot shows whether the estimated Year-40 mean continues to change substantially as more scenarios are added.

Tutorial/supporting task

A running average shows whether adding more scenarios still changes the estimated average outcome. Inspect it for the selected Year-40 indicator. The final average of the available runs is a useful comparison point, but it is still an estimate from this sample rather than the exact average of every possible future.

4.3 Convergence in the full uncertainty space

This repeats the sampling comparison with all selected uncertainties rather than only two visible inputs. Sobol needs more model runs as the number of uncertain inputs increases. If no scenario uncertainty is selected, each plan is evaluated once using its default path.

Tutorial/supporting task

Similar to the one above, just with greater uncertainty.

4.4 Quantifying coverage and discrepancy

Two-dimensional plots cannot show how well a sample covers all uncertain inputs at the same time. The numerical measures in this section provide an additional check of how evenly the full uncertainty space is represented.

Advanced Task (optional)

When many inputs vary at the same time, one two-dimensional plot cannot show whether the full range is well covered. Use the numerical checks below to assess coverage across the complete sample.

- Report the discrepancy score, which checks how evenly the points are spread; lower values indicate more uniform coverage.

- Report the average grid-coverage score across all pairs of inputs; higher values mean that more of the two-input combinations are represented.
- Discuss both sample size and design when comparing the diagnostics.
- Select the sampling methodology that performs best.

4.5 Establishing the working dataset

After comparing the sampling methods, the notebook creates one Latin Hypercube sample for the main baseline analyses. Reusing this dataset ensures that later results are based on the same set of sampled futures. The working dataset uses LHS.

Mandatory Task

This dataset is the common evidence base for the later outcome and sensitivity analyses, so it should be created once and reused. Create one reusable Latin Hypercube baseline dataset for Parts 5 and 6.

- Evaluate the baseline plan for all selected scenario inputs and store the results as df_baseline. This is the main baseline-results table, with one row for each sampled future.
- Report the value of N_SCENARIOS, which controls the requested sample size, together with the number of uncertain inputs and the selected outcome columns in the executed notebook.
- Check that all internal sample positions remain on the expected scale from 0 (low end) to 1 (high end), that all expected outcomes are present, and that shares and units are physically consistent.
- Do not resample separately for each later figure.

4.6 Visual overview of the working dataset

Before performing the detailed analyses, inspect the sampled inputs and resulting outcomes. These plots help identify unexpected values, strongly uneven distributions, or possible problems with the selected uncertainty ranges.

Tutorial/supporting task

These plots provide a first check of the sampled inputs and resulting baseline outcomes before more detailed analysis. Inspect the selected Year-40 inputs in physical units and the selected baseline outcomes. Distinguish means from medians and fixed values from sampled distributions.

4.7 Mapping outcomes onto the uncertainty space

This section shows a two-input projection of the full scenario dataset, where the other selected uncertainties also vary. It provides an initial visual indication of which parts of the uncertainty space may lead to favourable or unfavourable performance.

Tutorial/supporting task

This view links combinations of two uncertain inputs to the resulting Year-40 outcomes. Colour the two-input demonstration plane by selected Year-40 outcomes and identify combinations associated with favourable or unfavourable results. You can further extend this to visualize more pair-wise relationships.

5 Part 5: What range of outcomes is plausible?

5.1 Summarize the baseline outcome range

Here you describe how the baseline plan performs across all sampled futures. Before interpreting an indicator, use the notebook catalogue to check what it measures, its unit, and whether a higher or lower value is preferable. Pay particular attention to modal shares: strategic shares such as pt_share give longer trips more weight, while names ending in _share_trips count every modelled trip equally. These measures answer different questions and should not be mixed.

Mandatory Task

Summarise the baseline outcome space.

- Compare the Year-1 reference with the Year-40 median and spread for every selected indicator.
- Use the catalogue definitions when interpreting public-transport journey time, road delay, emissions, modal shares, and any optional counting-section indicators.
- Identify skewed or counter-intuitive results and trace them to the model mechanism before reporting them.
- Describe the sampled shapes as experiment results, not real-world forecast probabilities.

5.2 Year-40 outcome distributions and Year-1 references

A single average cannot show the full range or shape of the possible outcomes. The following distributions compare the Year-40 results with the Year-1 reference and make both change and uncertainty visible.

Mandatory Task

This comparison places the Year-1 reference beside the full Year-40 distribution so that both change and uncertainty are visible. Create and interpret the normalised distributional comparison.

- Show Year-40 distribution plots (violin plots) with quartiles and the Year-1 reference. If Year 1 is also uncertain, show its median and p10-p90 range; this range contains the middle 80% of the sampled results.
- Normalise every indicator with pooled Year-1 and Year-40 values and keep original units in the accompanying text or table. Normalisation changes the plotting scale, not the original result or its unit.
- Do not compare normalised heights as absolute magnitudes across different indicators.

6 Part 6: Which uncertain drivers explain the outcome spread?

This part asks which uncertain inputs are most closely connected to changes in the selected outcomes. Feature Scoring gives a quick ranking of inputs that help predict an outcome. Morris screening checks which inputs usually have a strong effect and whether that effect changes across the uncertainty space. Sobol analysis estimates how much of the outcome variation is linked to individual inputs and to interactions between them.

These methods are structurally different, but try to answer a similar question. You should use Feature Scoring to analyse the impact of different uncertain values and, if you want to complete the advanced tasks, compare such method with Morris and Sobol (both meant as optional, additional analyses).

6.1 Feature scoring: Year-40 outcomes

Feature Scoring provides a first ranking of the uncertain inputs associated with changes in the selected outcomes. It reuses the existing baseline dataset and therefore requires no additional transport-model evaluations.

Mandatory Task

A high score means that an input helps explain patterns in this sample. It does not mean that the input causes the same percentage of the outcome.

- Reuse df_baseline. In SENSITIVITY_INDICATORS, choose the outcomes to analyse; in DETAIL_INDICATOR, choose one of them for the more detailed charts and method comparison.
- Report the scores as relative rankings between inputs, not as percentages of outcome variation explained.
- Explain the top drivers using their model roles and do not generalise one outcome's ranking to other outcomes.
- Recognise that monetary inputs may be selected for later appraisal while having no effect on physical outcomes here.

6.2 Morris screening

Morris screening is meant as an optional method for identifying which inputs deserve more detailed analysis. It considers both the typical strength of each input’s effect and how much that effect changes across the uncertainty space.

Advanced Task (optional)

Morris screening is an optional next step that tests each input at several points in the uncertainty space. It helps narrow a larger set to the inputs that are most worth studying with the more demanding Sobol method.

- Compare the two Morris measures for each selected outcome. Mu-star shows the typical size of an input's effect; sigma shows how much that effect changes in different parts of the uncertainty space.
- A large sigma suggests that the effect may be non-linear or may depend on other inputs. It is not an error bar.
- Retain an input when it has a measurable effect on any selected outcome and document the rule used to reduce the set. Changing which Morris panels are displayed does not change the screening result.

6.3 Focused Sobol analysis on the Morris-reduced set

After Morris has reduced the number of inputs, Sobol analysis examines the retained inputs in greater detail. It distinguishes effects produced by individual inputs from effects that arise when inputs interact.

Advanced Task (optional)

Sobol analysis is the most detailed sensitivity method used here. It estimates how much of the variation in an outcome is associated with each retained input and with inputs acting together. The notebook runs it only on the smaller set retained by Morris to keep the computation manageable.

- Read the Sobol measures as follows: S1 is the contribution of one input acting on its own; ST is its full contribution, including interactions with other inputs; and S2 is the additional contribution of one specific pair of inputs acting together.

- Check whether the results stabilise as more runs are added, and do not over-interpret very small values close to zero.
- State clearly that these results assume every input excluded by Morris remains at its default value.

6.4 Compare rankings: Feature Scoring vs. Sobol

Feature Scoring and Sobol analysis measure sensitivity in different ways. Comparing their rankings helps determine whether they identify the same important inputs and where their conclusions differ.

Advanced Task (optional)

This comparison checks whether the quick Feature Scoring method and the more detailed Sobol method identify similar important inputs. Compare their rankings for the selected DETAIL_INDICATOR and explain agreements or disagreements. Different rankings are not automatically an error because the methods answer different questions.

7 Part 7: Compare static plans under the same scenarios

7.1 Build the paired plan-comparison dataset

Here the three fixed plans are evaluated under exactly the same sampled futures, which makes their results directly comparable. The notebook evaluates the same uncertainty draws for baseline (Stage 0), static1 (Stage 1 from the start), and static2 (Stages 1 and 2 from the start). Because every plan sees the same futures, differences can be attributed to the plan rather than to different scenario samples. These are annual operating snapshots, not adaptive plans. They show how each fixed configuration performs in a given year; they do not model a later decision to switch stages.

Mandatory Task

Create and verify the paired plan dataset df_plan. This comparison table contains one result for each plan within every sampled future, allowing like-for-like comparisons.

- Evaluate baseline, static1, and static2 with identical scenario draws.
- Verify explicitly that every scenario's uncertainty values match across all three plans.
- Report the number of scenario draws and total plan evaluations.
- Keep df_plan distinct from the baseline-only df_baseline dataset.

7.2 Outcome distributions by plan

Because all plans are evaluated under the same sampled futures, their outcome distributions can be compared directly. The following results show both typical plan performance and how widely each plan’s performance varies.

Mandatory Task

These distributions show the typical performance of each plan and how much its result changes across sampled futures. Compare the complete distributions for the selected indicators, not only their average values.

- Use the same vertical scale for Year 1 and Year 40 within each indicator.
- Report the exact Year-40 median and the p10-p90 range alongside the boxplots. This range contains the middle 80% of the sampled results.
- Identify outcomes for which plan effects are large relative to uncertainty and those for which distributions overlap.

- Use paired same-future changes relative to baseline to separate intervention effects from scenario differences.
- Do not select a preferred plan from one indicator alone.

7.2.1 Exact percentile summary by plan

The distribution plots provide a visual overview, while the percentile table reports the exact values. It can therefore be used to compare the median and uncertainty range of each plan more precisely.

Tutorial/supporting task

The percentile table gives exact central and low-to-high values for each plan. Use it when the numbers are more informative than estimating values from the boxplots.

7.2.2 Paired changes relative to the baseline

A paired change compares a plan with the baseline in exactly the same sampled future. This isolates the effect of the infrastructure choice from differences between futures. Subtract the baseline result from the plan result for each future. A negative value means a reduction, but whether that is good or bad depends on the indicator. Changes in modal share are reported in percentage points.

7.3 Parallel coordinates: plan profiles across futures

Looking at several indicators together can reveal trade-offs that are not visible in separate charts. The parallel-coordinates plot compares the overall performance profiles and uncertainty ranges of the three plans.

Tutorial/supporting task

Parallel coordinates show several indicators together, making plan profiles and trade-offs easier to see. Each vertical axis is scaled separately so indicators with different units can share one chart. Compare the plan lines and ranges, not the raw height of one indicator against another. A higher position is not automatically better; use the indicator definition to check the preferred direction.

7.4 Feature scores by plan

Infrastructure may change not only the outcome itself but also which uncertainties influence it most. This section compares the main drivers for the baseline, static1, and static2 plans.

Mandatory Task

Infrastructure can change which uncertainties drive an outcome, so feature scores are recalculated separately for each plan. Test whether the drivers of the selected Year-40 outcome change across baseline, static1, and static2.

- Use the same uncertainty columns and outcome definition for every plan.
- Identify stable drivers and parameters whose relative importance changes after infrastructure is introduced.
- Explain implications for evidence collection, monitoring, or design refinement.

8 Part 8: Which scenarios create vulnerabilities or opportunities? — Scenario discovery

Sensitivity identifies influential inputs; scenario discovery searches for combinations of their values associated with failure. The Patient Rule Induction Method (PRIM) describes these regions with rectangular bounds, which identifies a “cluster” of vulnerable scenarios (i.e. “bad” outcomes”).

8.1 Defining vulnerability/opportunity

Before searching for vulnerable scenarios, define exactly what counts as failure. In the notebook, VULNERABILITY_RULES lists the Year-40 indicator, comparison, and threshold for each rule. VULNERABILITY_MATCH chooses whether one failed rule is enough (any) or whether all selected rules must fail (all). The default example classifies a future as a failure when pt_share is below p.PT_SHARE_TARGET. This is a teaching benchmark, so you must justify it or replace it for your project.

Mandatory Task

Define and quantify the selected failure condition.

- Record every indicator, whether it must remain above or below its threshold, the threshold value, and whether any or all rules must fail.
- Check each indicator's definition and unit. The default pt_share weights public-transport trips by distance within the strategic trip population; it is not a simple count of trips.
- For baseline, static1, and static2, calculate the share of sampled futures that meet the chosen failure rule.
- Interpret that share only within this modelling experiment. It is not the real-world probability that failure will occur.

8.2 Where in the uncertainty space does baseline failure occur?

PRIM looks for a simple combination of input ranges in which baseline failure occurs unusually often. It describes that region as a box made from lower and upper limits for one or more inputs. Density tells you how many futures inside the box fail. Coverage tells you how many of all observed failures are captured by the box. Mass tells you how much of the full sample lies inside the box. A useful box should contain mostly failures while still capturing a meaningful share of them. In this notebook, a qualifying box must have at least 75% failure density and must narrow at least one input.

Mandatory Task

Use PRIM to investigate baseline failures.

- In PRIM_INPUTS, choose and justify the uncertain inputs used in the search. In PRIM_PLOT_INPUTS, choose up to two of them for the plot.
- Run PRIM only when the sample contains enough successful and failed futures for a meaningful comparison.
- Report every input range narrowed by PRIM, together with density, coverage, and mass.
- Plot the successful and failed futures with the selected box. State clearly if the full box also restricts inputs that are not visible in the two-dimensional plot.
- Translate the box into a planning narrative without assigning occurrence probability to it.

8.2.1 Interpreting the PRIM peeling trajectory

PRIM produces a sequence of possible boxes rather than one automatically correct result. The peeling trajectory shows how concentrating more failures inside the box can reduce the number of failures and sampled futures that the box captures.

Advanced Task (optional)

PRIM starts with all sampled futures and then removes small slices from the edges. Each step usually creates a smaller box with a higher concentration of failures, but it may also leave some failures outside. Explain this trade-off between density, coverage, and mass, and justify the selected box rather than automatically choosing the final step.

8.2.2 A clearer picture of the box

The numerical PRIM bounds can be difficult to interpret on their own. The following plot shows the discovered region using two selected inputs, while the accompanying results report any additional restrictions that are not visible in the plot.

Mandatory Task

This plot makes the selected PRIM ranges easier to see. Show every sampled future using the two display inputs, distinguish failure from success, and draw the selected box. Because the plot shows only two inputs, also report any additional input ranges that form part of the complete box.

8.3 Do the built stages still have vulnerabilities worth explaining?

Building an infrastructure stage may reduce failure without removing it under every sampled future. Repeating the same PRIM analysis for static1 and static2 shows whether clear vulnerability regions remain after the stages are introduced.

Mandatory Task

This check asks whether Stage 1 and Stages 1+2 still have a clear failure region under the same rules. Repeat the PRIM search for static1 and static2 when the sample contains enough successful and failed futures.

- Use the same rules, inputs, and density criterion as for baseline.
- If no qualifying box exists, report that result rather than forcing an unstable box.
- When the baseline box is shown only as a reference, label it clearly and recalculate its density and coverage for the displayed plan.

Exporting results for later phases

The notebook saves figures and tables as each analysis finishes, provided SAVE_OUTPUTS = True:

- Figures: figures/03_uncertainty_modeling/
- Tables: results/03_uncertainty_modeling/

The final export cells also save the experiment results, uncertainty schema, surrogate validation record and vulnerability summary under results/. These files document the assumptions and findings of Phase 3 and support later comparisons. Rerunning the notebook replaces the corresponding files.

Tutorial/supporting task

Run the final export cells and check that the saved results correspond to your intended uncertainty definitions, infrastructure configurations, indicators and vulnerability rules.

- Open the exported experiment table and verify the scenario count, plan names and selected indicators.
- Check that the saved figures and summaries support the findings presented in your report.
- Retain the notebook, configuration files and selected outputs together so that your analysis can be reproduced.
- Clearly distinguish discovered PRIM boxes from baseline reference boxes.

9 Required outputs for Phase 3

Mandatory Task

By the end of this phase, each group should retain and be able to explain:

- A justified uncertainty specification, including distributions, units and temporal behaviour.
- The surrogate training domain and relevant validation findings.
- The sampling design and size of the working scenario dataset.
- Selected Year-1 references and Year-40 outcome distributions.
- Feature-scoring results and their interpretation.
- Paired comparisons of the baseline and static plans, including percentile summaries and changes relative to baseline.
- A justified vulnerability definition and failure fractions by plan.
- PRIM findings where qualifying boxes exist, or a clear explanation when none qualifies.

A final preferred plan or claim of robustness is not required at this point. Phase 3 establishes how performance indicators vary across the selected uncertainty space, how static plans compare under common conditions, and where the configured target is vulnerable.

10 Connection to SIA

The activities carried out during Phase 3 support the systematic assessment of (deep) uncertainty and risk within the strategic planning process. The fixed infrastructure alternatives examined under nominal assumptions in Phase 2 are now tested across a wider range of plausible future conditions. This provides a stronger basis for developing adaptive pathways and completing the final appraisal in later phases.

In particular, the work supports the following planning activities:

- identifying assumptions and external developments that may influence infrastructure performance;
- defining transparent uncertainty ranges, distributions, and scenario-generation methods;
- testing whether evaluation indicators remain within acceptable levels under different future conditions;
- identifying the parameters for which stronger evidence, monitoring, or calibration may be required;
- comparing infrastructure alternatives under the same external conditions;
- recognising trade-offs between typical performance, uncertainty ranges, and vulnerability;
- describing combinations of uncertain conditions under which project objectives or planning targets may not be achieved; and
- preparing a documented evidence base for adaptive decision rules, monitoring indicators, and subsequent appraisal.

The uncertainty datasets, sensitivity rankings, static-plan comparisons, and vulnerability descriptions developed in this phase provide the quantitative basis for the adaptive-planning analysis in Phase 4 (Notebook 04). They should be interpreted together with project-specific evidence, stakeholder knowledge, and qualitative effects that are not represented in the model.